

The number of chickens is three times that of rabbits, and the total number of chicken legs exceeds the total number of rabbit legs by 20. Determine the respective quantities of chickens and rabbits.

Standard answer

This is a classic "chickens and rabbits in the same cage" problem, characterized by the given conditions of "multiple relationship of heads and difference of legs." Such problems can be solved using the grouping method.

According to the conditions, the number of chickens is three times that of rabbits. We group 3 chickens and 1 rabbit into one set. In each set, the difference between the number of chicken legs and rabbit legs is calculated as:

$$2 \times 3 - 4 = 2 \text{ (legs)}$$

That is, in each set, the number of chicken legs is 2 more than that of rabbit legs. To determine how many such sets correspond to the total difference of 20 legs, we calculate:

$$20 \div 2 = 10 \text{ (sets)}$$

Therefore, the number of rabbits is:

$$1 \times 10 = 10 \text{ (individuals)}$$

And the number of chickens is:

$$3 \times 10 = 30 \text{ (individuals)}$$

Answer: There are 30 chickens and 10 rabbits.

Answer 1

To solve this problem, we can formulate a system of equations to model the given conditions. Let C denote the number of chickens and R denote the number of rabbits. Based on the problem statement, we establish two equations:

1. The number of chickens is three times that of rabbits:

$$C = 3R$$

2. The total number of chicken legs exceeds the total number of rabbit legs by 20. Since each chicken has 2 legs and each rabbit has 4 legs, this condition translates to:

$$2C = 4R + 20$$

Next, we solve this system of equations. Substituting $C = 3R$ from the first equation into the second equation, we obtain:

$$2(3R) = 4R + 20$$

Simplifying the left-hand side:

$$6R = 4R + 20$$

Subtracting $4R$ from both sides to isolate the variable term:

$$6R - 4R = 20$$

$$2R = 20$$

Solving for R , we find:

$$R = 10$$

Using the value of $R = 10$ in the first equation $C = 3R$, we calculate the number of chickens:

$$C = 3 \times 10 = 30$$

Thus, the solution yields 30 chickens and 10 rabbits.

Answer 2

Based on the problem's conditions, we can directly leverage the quantitative relationship between chickens and rabbits to solve the problem. We know:

1. The number of chickens is three times that of rabbits.

2. The total number of chicken legs exceeds the total number of rabbit legs by 20. Each chicken has 2 legs, and each rabbit has 4 legs.

Let the number of rabbits be R . Then, the number of chickens is $3R$.

In terms of leg counts, the total number of chicken legs is $2 \times 3R = 6R$, and the total number of rabbit legs is $4 \times R = 4R$. According to the problem, since chicken legs exceed rabbit legs by 20, we establish the equation:

$$6R - 4R = 20$$

Simplifying this equation yields:

$$2R = 20$$

Solving for R , we find the number of rabbits: $R = 10$.

Since the number of chickens is three times that of rabbits, the number of chickens is $3 \times 10 = 30$.

In summary, there are 30 chickens and 10 rabbits. This solution process does not introduce unmentioned letters for equation-setting or unknown-variable solving, nor does it involve decimals or related operations.